



Validation Certificate

Validation certificate id: validation-certificate:ncd-asthma:asthma-cr1:modular-parity-2026-06-02
Model card id: model-card:ncd-asthma:pilot
Scenario card id: scenario-card:ncd-asthma:asthma-cr1:pilot
Model id: ncd-asthma
Scenario id: asthma_cr1
Validation status: pass
Certificate signature: 43f870e255f440cc24b262f1c411985c7cb114d786e3b3d00cef0de7f81df7aa

Certification Statement

This certificate records the asthma modular-structure parity validation. The current monolithic asthma CR1 model and the composed modular asthma CR1 model were run with the Rust runtime using the same scenario-materialised unified API data bundle.

Parity Result

The original run emitted 66,803 timestamped rows. The composed run emitted 66,803 timestamped rows. The result hash for both runs is 401c20690cfa2d93e91c6df78664f6d8297261386d8e1cba7e520f4ed7f80970. The byte-level comparison status is match.

Component Contracts

Component	Mode	Nodes	Links
demographic-substrate	compiled_together	3	6
asthma-epidemiology-core	compiled_together	6	12
tobacco-risk-factor	link_then_inject_bundle	38	41
clinical-low-dose-beclom	compiled_together	8	8
clinical-high-dose-beclom	compiled_together	6	5
clinical-oral-prednisolone	compiled_together	6	5
clinical-inhaled-short-acting-beta-agonist	compiled_together	6	5
burden-metrics	downstream_handoff	17	28
economic-metrics	downstream_handoff	3	3
resource-handoffs	downstream_handoff	8	24

Interpretation Boundary

This is a parity-lowering modular pilot. The demographic component records the intended contract with the existing botech demographic model, but this composed model still lowers to the embedded asthma demographic implementation so that exact parity can be proven before cross-model execution is introduced.